Facility Sustainability Review: Auditorium HVAC and Event Load Management

Document ID: DOC_BLD_014 | **Type:** FACILITY_REPORT | **Publication:** 2024-10-18 | **Institution:** VESIT (VESIT Green Campus, Chembur)

FACILITY SUSTAINABILITY REVIEW: AUDITORIUM HVAC AND EVENT LOAD MANAGEMENT

Document Reference: VESIT/FAC/AUD-HVAC/2024/01

Facility Code: BLD_AUD (Auditorium Building)

Operational Authority: Campus Events Directorate and Facilities (DEP_ADMN).

Location: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai.

1. FACILITY CHARACTERISTICS AND OCCUPANCY PROFILE

The Campus Auditorium (BLD_AUD) encompasses 2,800 m2 of built infrastructure featuring an 800-seat multi-tier main auditorium, stage wing facilities, sound engineering booths, green rooms, and a spacious pre-function entrance foyer. The building hosts major institutional milestone events, technical symposia, national conferences, cultural festivals, and placement drives.

Unlike standard academic classrooms that maintain predictable weekday schedules, BLD_AUD exhibits an intermittent high-demand electrical load profile. During major technical symposiums or full-capacity conferences, instantaneous cooling and lighting power demand surges up to 180 kW. However, the facility remains unoccupied for approximately 65% of standard calendar weekdays, creating acute operational challenges in energy management.

2. ENERGY CONSUMPTION ANALYSIS AND HVAC DYNAMICS

In 2024, BLD_AUD consumed 134,200 kWh of grid electricity (47.9 kWh/m2/year), resulting in 110,044 kg CO2e of Scope 2 emissions based on CEA emission factor EF_ELEC_GRID (0.82 kg CO2e/kWh):

- Space Air Conditioning: Accounts for 72.0% of total building consumption (96,624 kWh/year). Space cooling is provided by two 70-ton packaged water-cooled chillers (total 140 TR capacity).

- Stage Lighting and Audio-Visual Electronics: 18.0% (24,156 kWh/year).

- Auxiliary Motors, Ventilation Fans, and Foyers: 10.0% (13,420 kWh/year).

Prior to deploying IoT sub-metering under INIT_IOT_ENERGY_MGT in March 2024, telemetry logs revealed that facility staff frequently initiated chiller pull-down cycles five to six hours before rehearsals, running chillers across vacant halls. Furthermore, high-power stage lighting transformers drew 3.8 kW continuous standby power even during days of complete building closure.

3. OPERATIONAL OPTIMIZATIONS IMPLEMENTED

To eliminate wasted energy, the facilities team deployed three control protocols:

- Occupancy-Linked HVAC Pre-Cooling Algorithm: Implemented an automated control schedule calibrated to indoor wet-bulb temperatures. Pre-cooling is strictly limited to 45 minutes prior to hall occupancy, utilizing variable frequency drives (VFDs) on air-handling units to modulate airflow based on real-time CO2 ppm levels.

- Automated Standby Contactor Isolation: Programmable relays now fully disconnect stage lighting transformer substations when the master auditorium lighting console is powered off.

- Quantified Energy Conservation: These operational changes achieved a measured 16.4% reduction in building consumption, saving 22,000 kWh annually (18,040 kg CO2e avoided).

4. PROPOSED 50 KWP SOLAR BIFACIAL RETROFIT (INIT_SOLAR_PH2)

The Auditorium possesses an unobstructed curved perimeter roof covering 1,200 m2. Phase-2 solar feasibility modeling (DOC_ENG_004) proposes installing 50 kWp of lightweight bifacial photovoltaic modules on this roof. The installation will generate 71,000 kWh annually, satisfying over 52.9% of the Auditorium's annual electricity demand on-site and abating 58,220 kg CO2e annually, advancing institutional targets TRG_NET_ZERO and TRG_RENEWABLE while supporting SDG 7, SDG 11, and SDG 13.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

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